



Forecasting network attacks before they complete

VIGHNAX · a world model for SIH26153 (NTRO): AI-based Network Attack Forecasting from Network Traffic Data

PROBLEM

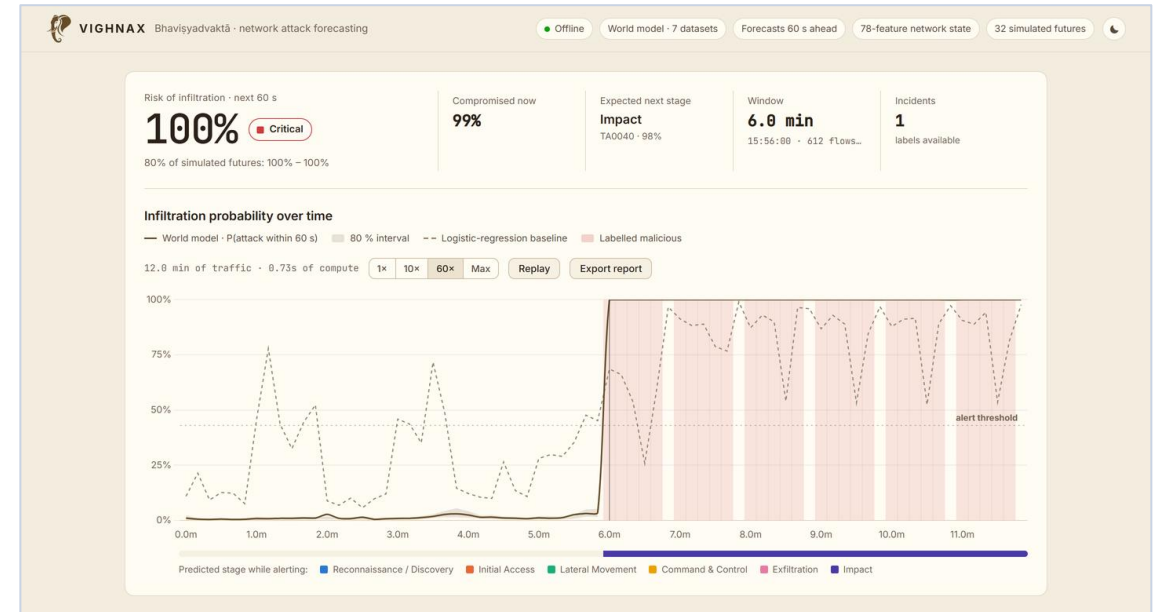
Intrusion detection labels each flow in isolation and fires after the fact, but an infiltration is a process that unfolds over time.

IDEA

Learn a world model of how the network's state evolves, $P(S_{t+1} | S_{\leq t})$, from flow and packet telemetry, then simulate it 60 s into the future.

OUTPUT, EVERY 10 S

$P(\text{attack within 60 s})$ with an uncertainty band, the next MITRE ATT&CK stage, and the features and hosts behind it.



Working prototype, fully offline: an unseen CIC-IDS2017 day turning critical as the DDoS begins.

54 / 201

attack onsets forecast before they began (baseline: 17)

0.70 vs 0.67

macro forecast AUROC on unseen data vs the baseline

7 of 7

PS datasets in one model, tested on unseen traffic

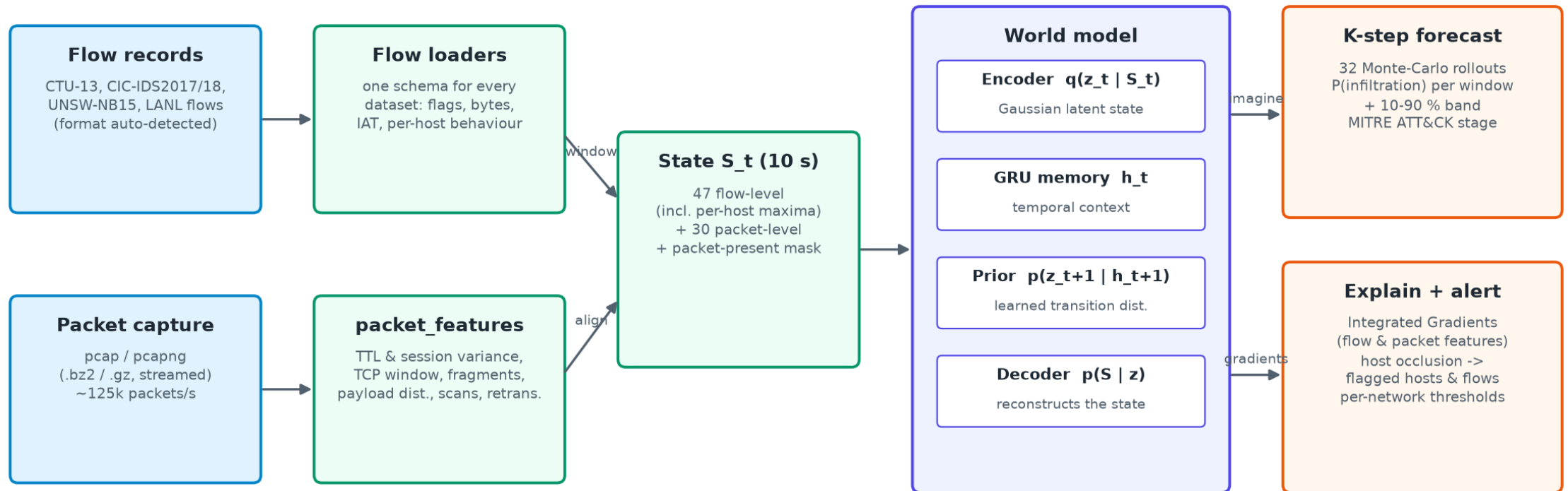
~8 ms

per 10 s window on a laptop CPU



Architecture: raw traffic to a forecast every 10 seconds

Flow records and/or packet captures $\rightarrow$ 78-dim network state S_t $\rightarrow$ world model $\rightarrow$ forecast, explanation, alert



Offline dashboard: live timeline, forecast band, stage mix, attributions, flagged hosts & flows, incident report | no cloud APIs

Flow level: 47 features

5-tuple statistics, flags, bytes, packets, duration, IAT, fwd/bwd ratios, per-host maxima so one bot is not diluted

Packet level: 30 features

TTL and session variance, TCP window, fragments, payload sizes, port-scan signatures, retransmissions

Ingestion

every PS dataset's native format, detected by content; PCAP / PCAPNG streamed at ~125k packets/s



World model: learn the dynamics, then imagine the future

RSSM-style latent state-space model (after World Models and Dreamer), trained on the attack timelines of all seven datasets

TRANSITION MODEL

encoder $q(z_t | S_t) \rightarrow$ GRU memory $h_t \rightarrow$ learned prior
 $p(z_{t+1} | h_{t+1}) \rightarrow$ decoder $p(S | z)$

TRAINED ON IMAGINED FUTURES

attack and ATT&CK-stage heads are scored on observed states and on K-step rollouts against the true future labels

FORWARD SIMULATION

32 Monte-Carlo futures $\times$ 6 steps (60 s): probability per window, 10–90 % band, stage mix

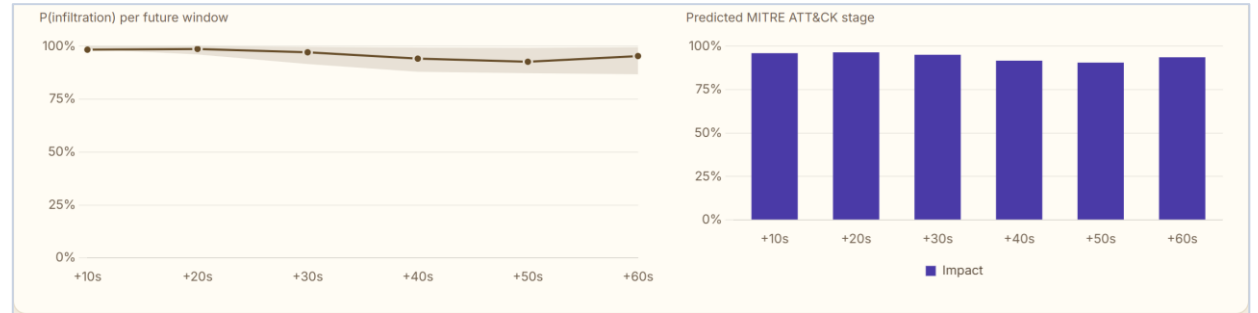
EXPLANATIONS

Integrated Gradients against a benign reference state; each host removed in turn to rank who drives the alert

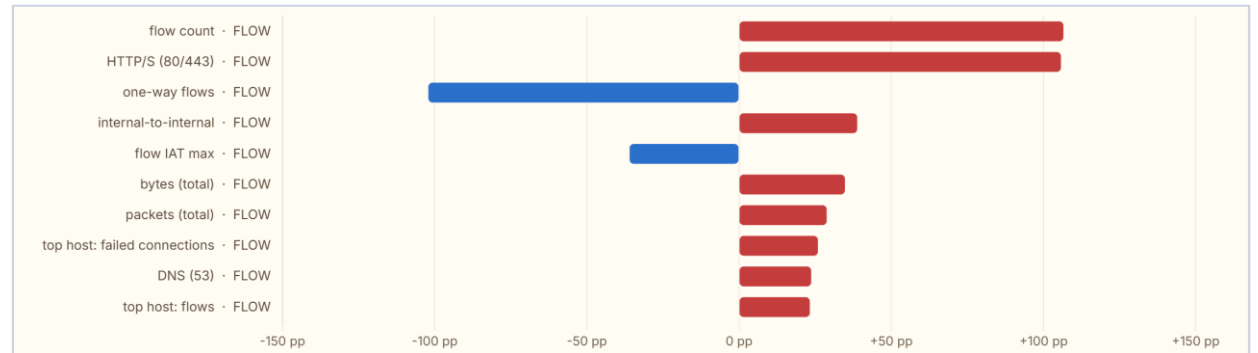
ROBUSTNESS

dataset-balanced training, modality dropout (CSV-only input), volume augmentation, per-site alert thresholds

WHAT HAPPENS NEXT · 32 futures, 60 s ahead



WHY · Integrated Gradients (red → compromise, blue → normal)



WHO · each host removed and the forecast recomputed

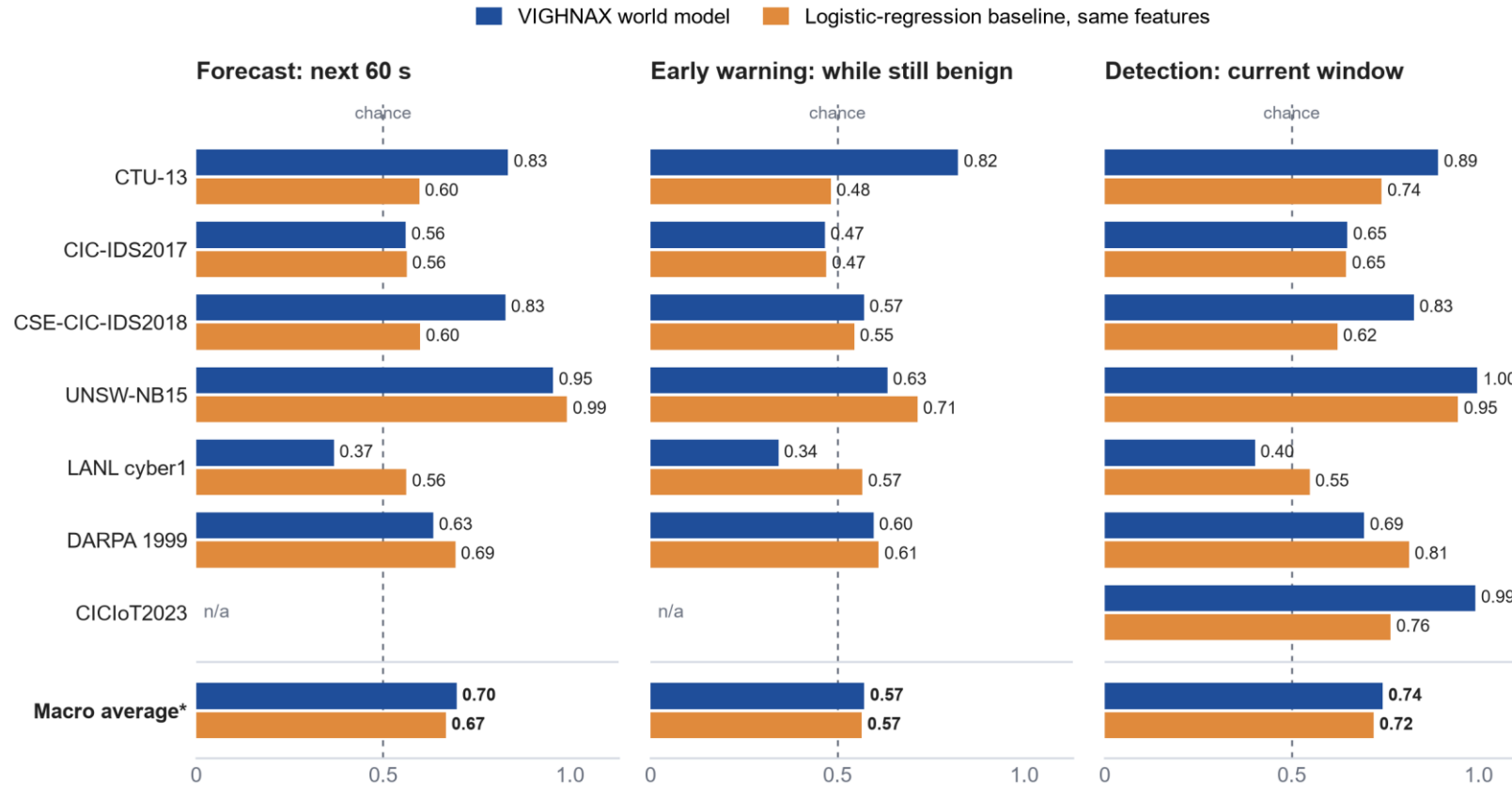
Host	Flows	Contribution to the forecast
172.16.0.1	598	+98.4 pp
108.161.188.128	5	+0.0 pp
104.97.71.5	2	+0.0 pp
104.24.112.68	1	+0.0 pp

Panels: real output at the DDoS onset of an unseen CIC-IDS2017 day.



Results: tested only on traffic the model never saw

AUROC, world model vs logistic regression on the same features · thresholds calibrated on validation only



0.5 = chance. Early warning = forecasting from windows that are still benign. CICIoT2023 episodes are composed, so only detection is scored. *Mean over the six temporal datasets.

54 / 201

attack onsets forecast before they began
(baseline: 17), 51–60 s ahead

WHERE WE LEAD

forecasting on CTU-13 and CSE-CIC-IDS2018; detection on 4 of 7 datasets

WHERE THE BASELINE LEADS

UNSW-NB15, LANL cyber1, DARPA 1999 (forecast); LANL cyber1 and DARPA 1999 (detection)

OPEN PROBLEM

on validation the world model leads everywhere; unseen days generalise less well



Demo, deployment and next steps

Runs offline on a laptop: run.bat / run.sh → http://127.0.0.1:5000, no internet or cloud needed

Demo

Drop a PCAP, flow CSV or .zip, or pick one of 8 bundled real captures. Timeline, forecast, stage mix, explanations and flagged hosts update live, window by window.

Deployment fit

Air-gapped and CPU-only; streams captures in bounded memory; SHA-256-verified weights; per-site alert thresholds; one-click incident report.



[Demo video](#)

YouTube · 2 minutes

Limits, stated plainly

Weak on LANL-type authentication attacks and behind the baseline on DARPA 1999; CICIoT2023 episodes are composed; thresholds need per-site calibration.

Next steps

Host-graph (GNN) state, online threshold calibration, a streaming NetFlow / IPFIX collector and SOC integration (syslog, STIX).



[Source code](#)

GitHub · weights · benchmark

Reproduce every number

fetch_datasets.sh → prepare_data.py / prepare_all.py → train.py → evaluate.py · one seeded config (configs/default.yaml, seed 42) · pytest suite